MIL-P-14634A
30 June 1974
SUPERSEDING
MIL-P-14634
28 Aug 1967

MILITARY SPECIFICATION

PANEL MARKER, RED CROSS, 4 FEET BY 4 FEET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type and size of red cross panel marker used by the Department of Defense.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL
V-T-285 - Thread, Polyester
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Shipping, Fiberboard
PPP-B-676 - Boxes, Set-Up
MILITARY
MIL-C-43256 - Cord, Polyester, Solid Braid
STANDARDS
FEDERAL
FED-STD-191 - Textile; Test Methods
FED-STD-751 - Stitch, Seam and Stitching
MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
DRAWINGS
THE INSTITUTE OF HERALDRY
5-1-357 - Panel Marker, Red Cross, 4' by 4'

/FSC 8345/

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(A miniature copy of drawing 5-1-357, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply:

COLOR ASSOCIATION OF THE UNITED STATES INC.
Standard Color Card of America, 9th Edition

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, NY 10016.)

3. REQUIREMENTS

3.1 Standard sample.- Unless otherwise specified (see 6.2), standard samples of all dyed fabric shall be furnished for matching color and colorfastness only.

3.2 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample of the finished panel marker shall be submitted or made available to the contracting officer or his authorized representative for inspection and tests as specified in 4.2. The approval of the first article authorizes the commencement of production but does not relieve the manufacturer of responsibility for compliance with all applicable provisions of this specification. The first article shall be manufactured in the same manner and by the same facilities to be used for the production items.

3.3 Material.- Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 Rip-stop fabric.- The fabric for the panel marker shall be a 1.5 ounce per square yard sail fabric woven from 70 ± 5 percent denier filament nylon with 104 ends and 100 picks per inch.

3.3.1.1 Weave.- The basic weave of the fabric shall be plain. The warp shall have 19 ends in each repeat with 17 ends weaving plain and 2 ends weaving as one. The filling shall have 20 picks on each repeat with 18 picks weaving plain and 2 picks weaving flat. There shall be a plain woven selvage on each side, 1/4, +1/8, -0 inch wide with 2 ends weaving as one.

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3.3.1.2 Color.- The cross portion of the panel shall be Old Glory Red, Cable No. 70180 in color and shall match the shade of the standard sample furnished (see 3.1). The field portion of the panel shall be White, Cable No. 70001 in color and shall match the shade of the standard sample furnished (see 3.1). When tested as specified in 4.4, the fastness of the dyed fabric to light, crocking, fresh and salt water shall be equal to or better than the standard sample. When no standard sample is available the dyed fabric shall show good fastness to light, crocking, fresh and salt water.

3.3.2 Reinforcement patches.- Reinforcement patches shall be fabricated from rip stop material specified in 3.3.1. The patches shall be White, Cable No. 70001 and shall match the shade of the standard sample furnished (see 3.1).

3.3.3 Tie cords.- The tie cords shall be fabricated from 3/32 inch in diameter, white or natural in color, polyester cord conforming to MIL-C-43256.

3.3.4 Thread, polyester.- Polyester thread shall conform to type I, any class of V-T-285 and shall be white. For type 301 stitching or type 304 stitching, size B shall be used for both the needle and the bobbin. For 401 stitching, size B shall be used for the needle and size AA shall be used for the looper.

3.3.5 Eyelets.- The anchoring eyelets shall be a commercial type rolled flange brass eyelet having a .280 inch nominal diameter hole.

3.3.6 Pins.- The anchoring pin shall be a commercial type 20d aluminum nail with a flat head.

3.4 Construction.- Panel markers shall be constructed in accordance with drawing 5-1-357 and the requirements specified herein.

3.4.1 Stitching.- All stitching shall be in accordance with FED-STD-751, using thread specified in 3.3.4. When a type 301 or type 304 stitch is specified there shall be 6 to 8 stitches per inch. When a type 401 stitch is specified there shall be 6 to 8 stitches per inch and the chain portion of the stitching shall not appear on the face side of the panel marker. All thread ends shall be trimmed.

3.4.1.1 Types 301 and 304 stitching.- Where a type 301 or a type 304 stitch is used, thread breaks shall be backstitched not less than 1 inch at each break. Except where ends are turned under in a hem or held down by other stitching, stitching shall be backstitched to prevent raveling. Thread tension shall be maintained so that there shall be no loose stitching and so the lock shall be embedded in the center of the material sewn.

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3.4.1.2 Type 401 stitching.- Where a type 401 stitch is used, thread breaks shall be over stitched not less than 1-1/2 inches at each break. Thread tension shall be maintained so that there shall be no loose stitching.

3.4.2 Setting of eyelets.- Eyelets specified in 3.3.5, shall be set by punching a hole through the material which is smaller than the barrel of the eyelet. The barrel of the eyelet shall then be forced through the punched hole and securely clinched without damaging or cutting the material.

3.4.3 Tie cords.- The tie cords shall be fabricated from polyester cord specified in 3.3.3. The cords shall have a free length (when measured from the edge of the panel) of 10 ± 1 inches. The cords shall be sewed to the panel marker using type 304 stitching.

3.4.4 Attaching of ancoring pins.- The aluminum anchoring pins shall be attached to the tie cords by means of a slip knot.

3.5 Marking.- Identification marking and any other marking required by the contract or purchase order, shall be imprinted or stencilled on the hem or reinforcement patch of the panel marker. The size of the lettering shall be sufficient to insure legibility. When tested as specified in 4.4, the marking shall show good fastness to weathering. The identification marking shall be as follows:

APPROVED NOMENCLATURE
STOCK NUMBER
NUMBER AND DATE OF CONTRACT

3.6 Workmanship.- The panel markers shall be clean, well made and shall meet the acceptable quality levels established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 First article inspection.- Inspection and testing of the first article shall be made on a completely finished panel marker for all provisions of this specification applicable to the end item examination and tests.

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4.3 Inspection for acceptance.- Inspection for acceptance shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.3.1 Inspection of components.-

4.3.1.1 Testing of components.- Testing of components shall be in accordance with table I and applicable subsidiary specifications and standards to the extent applicable. The method of testing specified in FED-STD-191, wherever applicable and as listed in table I shall be followed. All requirements are applicable to the sample unit. Unless otherwise specified in subsidiary specifications, sampling shall be in accordance with the following:

Lot Size	Sample Size
800 or less	2
801 thru 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The unit for expressing lot sizes and sample unit for testing each component shall be in table I.

TABLE I Component testing.-

Sample unit	Component and lot expressed in terms of	Characteristic	Rqm't para	Test method	Rqm't per S/U	#determ per S/U	Results reported as
1-1/2 yards	Cloth, nylon rip stop (linear yd)	Material ident	3.3.1	1/	x	1	Pass or fail
		Denier		1/	x	1	Pass or fail
		Weight oz/sq yd		5041	x	5	Avg to nearest 0.01 oz
		Texture	3.3.1 & 3.3.1.1	Visual	x	1	Pass or fail
		Weave	3.3.1.1	Visual	x	1	Pass or fail
		Colorfastness to:					
		Light	3.3.1.2	5660			
		Crocking	3.3.1.2	5651			
		Fresh water	3.3.1.2	5630			
		Salt water	3.3.1.2	5632			
		Marking 2/	3.5	5671	x	1	Pass or fail
2 Eyelets	Eyelets (ea)	Material ident	3.3.5	1/	x	1	Pass or fail
		Dimensions	3.3.5	Visual	x	1	Pass or fail
1 pin	Pins	Material ident	3.3.6	1/	x	1	Pass of fail

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1/ Acceptable under certification of compliance procedures.
2/ The nylon rip stop cloth sample unit shall be imprinted or stencilled with the identification indicated in 3.5, using the same method and material to be used for the production items.

4.3.2 In-process inspection.- Inspection of the subassemblies shall be made at any point of the manufacturing process to determine whether operations or assemblies are accomplished as specified on the applicable drawing or as specified herein. The Government reserves the right to exclude from consideration for acceptance any assembly for which in-process inspection has indicated non-conformance.

4.3.3. Inspection of the end item.-

4.3.3.1 Visual examination.- Defects found during the examination shall be classified in accordance with 4.3.3.1.1 and 4.3.3.1.2. Lots shall be inspected in accordance with 4.3.3.1.3. The unit of product for these examinations shall be one completely fabricated panel marker.

4.3.3.1.1 Examination for defects in material, design, construction and workmanship.-

Examine	Defect	Classification	
		Major	Minor
Raw edges	When securely caught in stitching:		
	-more than 1 inch, but not more than 3 inches.	-	x
	-more than 3 inches.....	x	-
	NOTE: Raw edges not securely caught in stitching shall be classified as an open seam.		
Seam and stitch type	Wrong seam or stitch type:		
	-seriously affecting serviceability.....	x	-
	-affecting serviceability, but not seriously..	-	x
Run-offs (see open seams)			
Stitch tension	Loose, resulting in a loosely exposed bobbin or top thread; tight, resulting in excessive tightness of fabric on seams.....	-	x
Stitches per inch	One stitch less than the minimum specified.....	-	x
	Two or more stitches less than the minimum specified.....	x	-
	One stitch in excess of maximum specified.....	-	x
	Two or more stitches in excess of maximum specified.....	x	-
Stitching margins; gage on double needle stitching	Under or over specified width by more than 1/16 inch.....	-	x

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Examine	Defect	Classification	
		Major	Minor
Thread breaks	Overstitched less than 1-1/2 inches on stitch type 401; backstitched less than 1 inch on stitch type 301	-	x
	NOTE: Thread breaks not backstitched or not overstitched shall be classified as open seams.		
Stitching ends	Not backstitched (except where ends are held down by other stitching or turned under in a hem)	x	-
Components and assembly	Any required component, any required operation omitted or not as specified (except as otherwise specified herein):		
	-seriously affecting serviceability	x	-
	-affecting serviceability, but not seriously	-	x
	Needle chews expected to develop into a hole	-	x
	Seam badly pleated or puckered	-	x
Stitching	One or more required rows of stitching omitted	x	-
Fabric	Holes (except punched holes), cut or tear	x	-
	Exposed punched hole, i.e., not covered by attached parts	x	-
	Color of base cloth not as specified	x	-
	Color of base cloth off shade	-	x
	Fabric defects:		
	-seriously affecting serviceability	x	-
	-affecting serviceability, but not seriously	-	x
Tie cord	Edges frayed or scalloped; misplaced failing to serve intended purpose	-	x
	Finished length less than specified by more than one inch	-	x
	Tie cord attached to face of panel marker	-	x
Eyelet	Broken, cracked or malformed; seriously affecting serviceability	x	-
	Broken, cracked or malformed; affecting serviceability, but not seriously	-	x
	Size or type not as specified:		
	-seriously affecting serviceability	x	-
	-affecting serviceability, but not seriously	-	x
	Missing	x	-
	Clinched excessively tight, cutting fabric or insecurely clinched	x	-

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Examine	Defect	Classification	
		Major	Minor
Pins	Not aluminum	-	X
	Bent, broken, malformed or otherwise defective, affecting serviceability	-	X
	Point blunted	-	X
	Not correct size	-	X
	Not attached to cord with slip knot	-	X
Seams and stitching	Open seam on stitch type 301:		
	-more than 1/2 inch, but not more than 1 inch.	-	X
	-more than 1 inch	X	-
	Any open seam on stitch type 401, except run-offs:		
	-3/4 inch or more	X	-
	-more than 1/4 inch but less than 3/4 inch	-	X
	Run-offs on stitch type 401:		
	-more than 1/2 inch, but not more than 1 inch.	-	X
	-more than 1 inch	X	-
	Chain portion of 401 stitch type appears on face side	-	X
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped stitches or run-offs occur. On double stitched seams, a seam shall be considered open when either one or both sides of the seam are open.		
Hems	Twisted or pleated	-	X
	Hem uneven	-	X
	Hem on wrong side of panel marker	-	X
	Constructed with a single turn under when a double turn under is required	-	X
Reinforcement patches	Any reinforcement misplaced, failing to reinforce eyelet fully	-	X
	Color of reinforcement patch not white	-	X
Marking	Omitted, illegible, incorrect or misplaced	-	X
Cleanness	Noticeable grease or oil stains; thread ends not trimmed throughout	-	X

4.3.3.1.2 Dimensional defects.- Any dimension that is not within the specified tolerance shall be classified a defect.

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4.3.3.1.3 Inspection levels and acceptable quality levels (AQL's).- The inspection levels and the acceptable quality levels expressed in defects per hunderd units shall be as follows:

	Inspection level	AQL Major	AQL Total
For defects applicable to 4.3.3.1.1	II	4.0	10.0
For defects applicable to 4.3.3.1.2	S-4	(one class)	2.5

4.3.4 Examination of preparation of delivery requirements.- An examination shall be made to determine that packaging, packing and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Shipping containers fully prepared for delivery shall be examined for defects of closure listed below. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, any component damaged affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, inadequate stapling.
Contents	Number per container is more or less than required.

4.4 Tests.- Testing shall be in accordance with FED-STD-191. Fastness to weathering shall be tested in accordance with 5671. Fastenss to fresh water shall be in accordance with 5630. Fastness to salt water shall be tested in accordance with 5632. Fastness to crocking shall be tested in accordance with 5651 and fastness to light shall be tested in accordance with 5660.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Unit package.- One panel marker (with anchoring pins attached) shall be neatly folded and placed in a snug-fitting transparent polyethelene bag approximately 6 inches wide and 10 inches deep. The polyethelene film shall be a minimum of 0.004 inch thick. All seams and closures shall be effected by heat sealing. Prior to or during the heat sealing operation excess air within the bag shall be expelled.

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5.1.1.2 Intermediate package.- Ten panel markers prepared as specified in 5.1.1.1, shall be packaged in a snug-fitting paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. Box closure shall be in accordance with the appendix of the container specification.

5.1.2 Level C.- Panel markers shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving station. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A.- Panel markers, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to class 2, style 4 of PPP-B-621 or overseas type, style A, B, I or J of PPP-B-601. Box closure and reinforcing shall be in accordance with the appendix of the applicable container specification.

5.2.2 Level B.- Panel markers packaged as specified in 5.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF or SF, class domestic, grade 275, style FOL of PPP-B-636. Closure shall be in accordance with the appendix of the container specification. The weight of contents in each container shall not exceed the size and weight limitations established by the container specification.

5.2.3 Level C.- Panel markers packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination for the lowest transportation rates for such supplies. Containers shall be in accordance with the rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking requirement in the contract or purchase order, unit packages, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Panel markers covered by this specification are intended to be used for ground to air identification of emergency helipads.

6.2 Ordering date.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a standard sample will not be furnished (see 3.1).

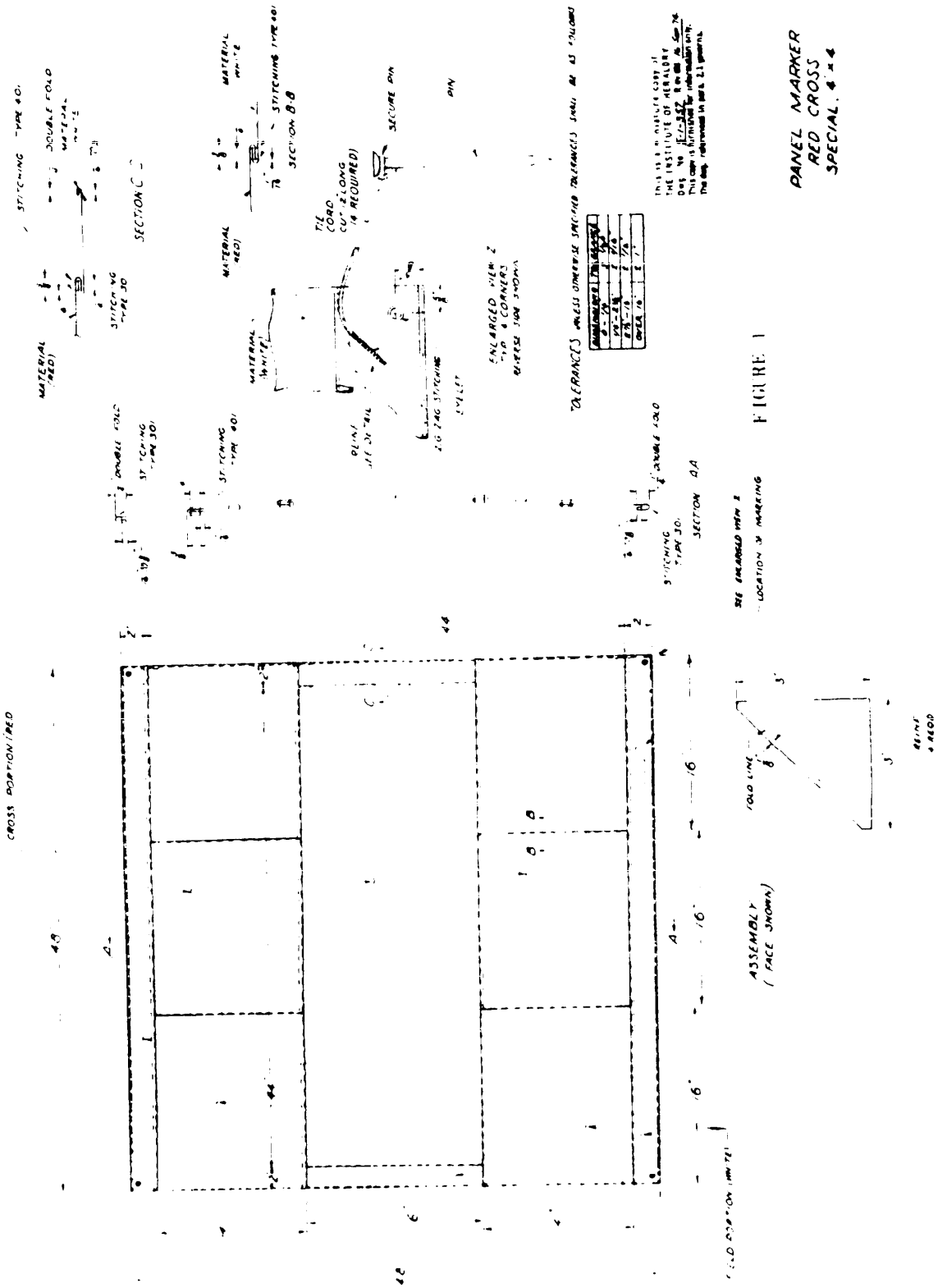
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(c) When a first article is not required (see 3.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Custodians:
Army - IH
Air Force - 82
Navy - SA
Review Activities:
Air Force - 82
Navy - MC, SA
Other - DP
User Activities:
Army - SM
Project No. 8345-0080

Preparing Activity:
The Institute of Heraldry,
United States Army

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-P-14634A Panel Marker, Red Cross, 4 Feet by 4 Feet

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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THE INSTITUTE OF HERALDRY, U.S. ARMY
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